Question			Marking details	Marks available					
				AO1	AO2	AO3	Total	Maths	Prac
5	*(a)*		[Induced] emf (or pd or voltage) equal to (or proportional to) the rate of change (or cutting) of flux [linkage] (1) The emf [tends to] oppose the <u>change</u> [to which it is due] (1)	2			2		
	(b)		**Indicative content:** Flux in pipe(s) changes or flux cut by pipe(s) Emf induced in pipe(s) Pipe P current cannot flow / incomplete circuit Pipe Q current flows [Current] opposes motion (change) …. …..(since) magnetic field set up due to (induced) current Uniform acceleration - no opposing force or gravity only **Additional useful points –** Terminal velocity - magnetic force equal (and opposite) to weight Slow velocity - due to strong magnet or small resistance or large current GPE converted to internal energy or electrical energy or electromagnetic energy Copper is not magnetic		6		6		

Question			Marking details	Marks available AO1	AO2	AO3	Total	Maths	Prac
			5-6 marks Comprehensive list of observations made. *There is a sustained line of reasoning which is coherent, relevant, substantiated and logically structured.* **3-4 marks** Some reasonable observations made. *There is a line of reasoning which is partially coherent, largely relevant, supported by some evidence and with some structure.* **1-2 marks** Limited observations made. *There is a basic line of reasoning which is not coherent, largely irrelevant, supported by limited evidence and with very little structure.* **0 marks** *No attempt made or no response worthy of credit.*						
	(c)		mgh **or** $mc\Delta\theta$ used **or** implied (i.e. E_p = 2.35 J) (1) Use of $V = Ah$ (1) Use of $m = \rho V$ (1) (i.e. m = 0.056 kg) Answer = 0.109 [K] (1)		4		4	4	
			Question 5 total	**2**	**10**	**0**	**12**	**4**	**0**